Unsupervised Image-Adapted Local Fisher Discriminant Analysis to Reduce Hyperspectral Images Without Ground Truth

Rania Zaatour, Sonia Bouzidi, and Ezzeddine Zagrouba

***Abstract*— Local Fisher discriminant analysis (LFDA) is a feature extraction technique that proved efficient to reduce several types of data and succeeded to outperform many state-of-the-art methods. However, due to its supervised nature, LFDA's efficiency depends on the available labeled samples and declines dramatically when the latter are very few. Hence, we assume that we cannot resort to LFDA to reduce unlabeled data. In this article, we studied to what extent this assumption is true and questioned the possibility of using LFDA to reduce hyperspectral images (HSIs) with no available ground truth. To study the real impact of the labeled information on LFDA's performance, we replaced the costly expert-made ground truth by different sets of labeled samples that are generated based on the image's offered spectral and/or spatial information, with no prior knowledge of the captured scene nor of its classes. Our proposed sets proved able to guide LFDA in extracting relevant discriminating features. This proved that LFDA does not depend only on expert-made labeled information and led us to define the unsupervised image-adapted LFDA (uiaLFDA) that can properly reduce an HSI without requiring its ground truth. To do so and to replace the ground truth that LFDA usually requires to reduce an HSI, uiaLFDA defines its own set of labeled samples by simply gridding the image into cells where each cell is considered a class. Our experiments ran on three HSIs proved that uiaLFDA is as efficient as LFDA and, even better, in reducing unlabeled HSIs.**

***Index Terms*— Dimensionality reduction, hyperspectral images (HSIs), local Fisher discriminant analysis (LFDA).**

I. Introduction

WHEN dealing with a voluminous data set, it is often a must to reduce its dimensions. This preprocessing step lessens the heavy computational burden caused by the data's volume and minimizes the data's redundancy and correlations without losing its main information.

Dimensionality reduction techniques fall into one of the following two categories: 1) feature selection methods that reduce a given data by selecting its most representative features and 2) feature extraction methods that project the data into a smaller subspace where the reduced features represent to the best the original voluminous data. In both categories, we find unsupervised, supervised, and semisupervised techniques.

In this article, we are interested in a supervised feature extraction technique called local fisher discriminant analysis (LFDA) [1]. This latter reduces a voluminous data set by mapping it to a lower dimensioned space where the intraclass variance is minimized, the interclass variance is maximized, and the local structure of the data is conserved.

Through the last decade, LFDA proved to outperform many state-of-the-art dimensionality reduction techniques in reducing different types of data. In fact, it helped recognize spoken emotions [2], facial expressions [3], ear images [4], faces [5], palm prints [6], range deception jamming [7], spoken language [8], and actions [9]. Besides, it was used for person reidentification [10] and for fault diagnosis [11]. Moreover, in the remote sensing field, LFDA was employed to reduce the dimensionality of hyperspectral images (HSIs) [12]–[14].

Even though LFDA proved its versatility and efficiency in several contexts, it suffers from two major problems.

On one hand, LFDA computes its transformation matrix in a greedy classwise fashion that requires a lot of memory and time, mainly in the case of higher dimensioned data sets with larger classes. To surpass this problem, several solutions were proposed in the literature, such as applying LFDA to a smaller wisely selected subset of features instead of the original voluminous data [15] or using sparse matrices and faster searching algorithms [9]. More recently, we introduced the class-adapted LFDA (caLFDA) [14] that is able to reduce any given labeled data—regardless of its dimensions and of the available memory of the used machine—by considering large classes as the sets of smaller subclasses and by processing the latter instead of the larger classes.

On the other hand, LFDA's second flaw is due to its supervised nature, which causes its performance to depend on the available expert-made labeled samples and to decrease drastically when the latter are very few. To overcome the lack of this supervised information, several semisupervised versions of LFDA were proposed, such as: 1) SELF [16] and its enhanced versions [17]–[19] that linearly combine LFDA with the unsupervised principal component analysis (PCA) [20]; 2) S2LFDA [21] that uses the reconstruction probability class to assign to each unlabeled sample a probability of belonging to a given class; 3) the enhanced LFDA [22] where a signal perturbation method generates more representative training

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samples; or 4) the transductive LFDA [23] that imposes smoothness constraints to effectively exploit the unlabeled samples in the learning process.

Regarding the aforementioned problems and existent solutions, we know that: 1) having more available labeled samples ameliorates LFDA's performance; 2) in case, many available labeled samples form larger classes that yield heavier memory consumption, we can rely on one of LFDA's optimized solutions, such as caLFDA; 3) having very few labeled samples ruins LFDA's performance but we can rely on one of its semisupervised versions; and 4) in case of total lack of expert-made labeled samples, we *a priori* assume that we cannot use the supervised LFDA.

However, in this last case, to what extent are our assumptions true? Is LFDA really incapable of effectively reducing a given data in the absence of its expert-made labeled samples? Is there a way to replace the latter?

In this article, in order to answer these questions, we propose to study the relevance of the expert-made labeled samples to LFDA and to inquire about their impact on its performance. Particularly, we are interested in leading this article on the highly dimensioned HSIs that, due to their expensive and time-consuming labeling process, are abundantly available yet rarely labeled, and if so, their ground truth would only show very few expert-made labeled samples.

In order to determine to what point LFDA depends on the expert-made ground truth, we, hereby, replace this latter by a set of class labels that are automatically generated without requiring human expertise nor prior knowledge of the data. To do so, and since HSIs offer very rich spatial and spectral information, we propose to exploit one or both of this information to define region maps that we later feed to LFDA instead of the costly and hard to provide ground truth.

In view of the outcome of this article, we, hereby, propose the unsupervised image-adapted LFDA (uiaLFDA) that extends the limited supervised LFDA to properly reduce the challenging highly dimensioned HSIs by replacing their unavailable ground truth by a simple yet effective gridded region map. In fact, we are calling uiaLFDA: 1) "unsupervised" as it does not require an expert-made ground truth nor a prior knowledge of the captured scene and 2) "image-adapted" since it uses the information offered by the image to define its own set of labeled samples. By doing so, unlike LFDA, uiaLFDA is not constrained by the number of samples for which we know the true class of land cover.

The rest of this article is structured as follows. Section II gives brief reminders of LFDA and caLFDA and details the motivation of this article. Next, Section III studies the influence of the labeled information on LFDA by replacing the costly ground truth by several labeled region maps that are generated based on the image's offered information and by studying the impact each of them has on the efficiency of LFDA. Based on the outcome of this article, Section IV introduces our proposed uiaLFDA, tunes its parameters, and puts it to the test using three HSIs. Finally, Section V wraps this article with some conclusions and perspectives.

II. Background and Motivation

A. LFDA

LFDA [1] was defined by combining the supervised FDA [24] and the unsupervised locality preserving projection (LPP) [25].

Like FDA, LFDA seeks a transformation that maximizes the between-class variance and minimizes the within-class one. However, unlike FDA, LFDA can effectively handle multimodal data (i.e., made of several clusters or containing outliers) since, like LPP, it preserves the locality of the samples in the original space. Consequently, LFDA projects the data in an embedding space where different classes are separated without inevitably making far samples of the same class close.

In order to reduce a set of n d-dimensional samples representing c classes, $X = \{x_1, \ldots, x_n \mid x_i \in \mathbb{R}^d\}$, given the set of their respective class labels, $Y = \{y_1, \ldots, y_n \mid y_i \in \{1, \ldots, c\}\}$, LFDA starts by computing LPP's affinity matrix A, as in the following equation:

$$A_{i,j} = \exp\left(-\frac{\left\|x_i - x_j\right\|^2}{\left\|x_i - x_i^{(k)}\right\|\left\|x_j - x_j^{(k)}\right\|}\right) \tag{1}$$

where $x_i^{(k)}$ is the kth nearest neighbor of x_i, and $k = 7$ is a heuristic that altogether works well according to [26].

In view of the proximity A measured based on the samples' local scaling, LFDA computes the local between-class and the local within-class weight matrices, as in the following equations:

$$W_{i,j}^{lb} = \begin{cases} A_{i,j}\left(\frac{1}{n} - \frac{1}{n_l}\right), & \text{if } y_i = y_j = l, \\ \frac{1}{n}, & \text{if } y_i \neq y_j \end{cases} \tag{2}$$

$$W_{i,j}^{lw} = \begin{cases} \frac{A_{i,j}}{n_l}, & \text{if } y_i = y_j = l, \\ 0, & \text{if } y_i \neq y_j \end{cases} \tag{3}$$

where n_l is the number of samples in class l ($\sum_{l=1}^{c} n_l = n$).

Given the $n \times n$ weight matrices, LFDA computes the local between-class scatter matrix S^{lb} and the local within-class scatter matrix S^{lw}, as in (4) and (5), respectively

$$S^{lb} = \frac{1}{2}\sum_{i,j=1}^{n} W_{i,j}^{lb}(x_i - x_j)(x_i - x_j)^\top \tag{4}$$

$$S^{lw} = \frac{1}{2}\sum_{i,j=1}^{n} W_{i,j}^{lw}(x_i - x_j)(x_i - x_j)^\top. \tag{5}$$

Finally, to reduce the data into $r \leq d$ dimensions, LFDA defines its transformation matrix T_{LFDA}, as in the following equation:

$$T_{\text{LFDA}} = \underset{T \in \mathbb{R}^{d \times r}}{\arg\max}\left[tr\left(\frac{T^\top S^{lb} T}{T^\top S^{lw} T}\right)\right]. \tag{6}$$

Equivalently, by solving the generalized eigenvalue problem of (7), we can obtain an analytic form of T_{LFDA}, as in (8)

$$S^{lb}\varphi = \lambda S^{lw}\varphi. \tag{7}$$

$$T_{\text{LFDA}} = \left(\sqrt{\lambda_1}\varphi_1 | \sqrt{\lambda_2}\varphi_2 | \ldots | \sqrt{\lambda_r}\varphi_r\right) \tag{8}$$

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where $\{\varphi_k\}_{k=1}^{r}$ are the normalized eigenvectors associated with the generalized eigenvalues $\{\lambda_1 \geq, \ldots, \geq \lambda_r\}$.

Since computed in a classwise manner, when trying to reduce a highly dimensioned data set containing very large classes on commodity hardware, LFDA either fails to run or consumes all the available limited RAMs and blocks the use of the machine. To solve this problem, we proposed the caLFDA [14].

B. caLFDA

Sugiyama's implementation of LFDA[1] starts by initializing the scatter matrices, S^{lb} and S^{lw}, to zeros. Then, considering one class $l \in [1, \ldots, c]$ at a time, LFDA measures the distances between its n_l samples and updates S^{lb} and S^{lw}.

However, when using commodity hardware, resorting to LFDA to reduce a highly dimensioned data set that contains very large classes involves computing very large $n_l \times n_l$ affinity matrices, which might consume all the available limited memory or even require more.

To overcome this memory issue and make the most of LFDA using limited hardware, we proposed the caLFDA [14], by constraining LFDA to: 1) consider large classes as a set of smaller subclasses and 2) process these handleable subclasses instead of the larger problematic class.

To do so, caLFDA defines a limit value $n_{\max}$ as the maximum number of samples per class it can handle, divides every large class containing more than $n_{\max}$ samples into smaller subclasses containing at most $n_{\max}$ samples each, and computes the affinity matrices of these smaller subclasses and updates the scatter matrices. Hence, if the data to be processed do not contain large classes made of more than $n_{\max}$ samples, caLFDA is reduced to LFDA. Note that formula to compute $n_{\max}$ based on the available resources and on the data to be processed can be found in [14].

Considering that caLFDA uses a class-adapted set of class labels Y^{ca}, where the classes are limited to at most $n_{\max}$ samples, we can say that the main difference between LFDA and caLFDA resides in the way weight matrices are defined. In fact, caLFDA defines its weight matrices as in the following equations:

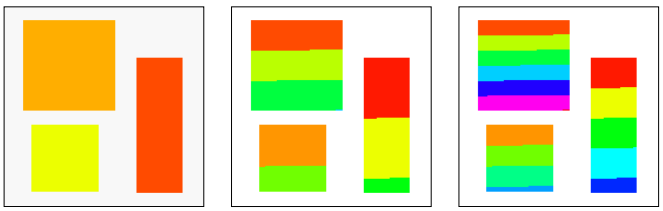
$$W_{i,j}^{lb} = \begin{cases} A_{i,j}\left(\dfrac{1}{n} - \dfrac{1}{n_l^{ca}}\right), & \text{if } y_i^{ca} = y_j^{ca}, \\ \dfrac{1}{n}, & \text{if } y_i^{ca} \neq y_j^{ca} \end{cases} \tag{9}$$

$$W_{i,j}^{lw} = \begin{cases} \dfrac{A_{i,j}}{n_l^{ca}}, & \text{if } y_i^{ca} = y_j^{ca}, \\ 0, & \text{if } y_i^{ca} \neq y_j^{ca} \end{cases} \tag{10}$$

where y_i^{ca} is the class-adapted label of the sample x_i (i.e., $y_i^{\text{ca}} = y_i$ if $n_l \leq n_{\max}$) and n_l^{ca} is the size of class or subclass l (i.e., $n_l^{\text{ca}} = n_l$ if $n_l \leq n_{\max}$; otherwise, $n_l^{\text{ca}} \leq n_{\max} < n_l$).

Once the weight matrices defined, caLFDA proceeds to compute the scatter matrices exactly as does LFDA in (4) and (5). Finally, caLFDA seeks a transformation matrix that, as LFDA's (6), maximizes the between-class variance to well separate the original handleable classes and the newly defined subclasses and minimizes the within-class variance to make the classes and subclasses more compact. Consequently, compared with LFDA, caLFDA will still discriminate the classes in the data while making the samples of large classes more or less close depending on the defined subclasses.

[1]Available online at www.ms.k.u-tokyo.ac.jp/software.html#LFDA



(a) (b) (c)

Fig. 1. (a) Simulated ground truth with 171×171 pixels, and caLFDA's set of labeled samples defined with (b) $n_{\max} = 2000$ and (c) $n_{\max} = 1000$.

Using a thorough evaluation protocol, [14] proved that caLFDA overcomes LFDA's problem and can efficiently reduce highly dimensioned data sets on commodity hardware. Moreover, in considerably less time, caLFDA can extract features that are as good as the ones LFDA extracts, which yielded very similar or even better classification results.

Finally, it is worth mentioning that this sort of divide and conquer solution can be easily parallelized and distributed to reduce very highly dimensioned data sets using the cloud computing architectures, as proposed in [13].

C. Motivation

When introduced in [14], caLFDA was applied to reduce HSIs that come with a set of labeled samples called the ground truth. This latter is the result of a very long and expensive process that requires many resources to: 1) identify the different land cover and land use classes in the captured scene by collecting samples from small surfaces *in situ*; 2) label the pixels corresponding to the collected samples; and 3) generalize for the larger unlabeled surfaces.

For supervised HSI processing, the ground truth is mandatory since it provides the supervisory and prior knowledge of the data to be processed. However, in the case of a ground truth where classes are too large, caLFDA does not use this ground truth as it leads to its discriminating process. Instead, it adapts it to create a class-adapted set of labeled samples where problematic large classes are limited to $n_{\max}$ samples and where samples that originally belong to the same large class might belong to different subclasses.

To properly illustrate this, let us consider Fig. 1 that compares a simulated ground truth to two of caLFDA's sets of labeled samples defined according to two values of $n_{\max}$ and proves how, by considering the new subclasses, caLFDA treats some samples of the same real class as if belonging to different classes.

Theoretically speaking, and with regards to the definition of supervised processing, this behavior is inconsistent and divergent from the supervised context. Yet still, caLFDA succeeded to extract relevant features. What is more confusing is that with its class-adapted sets of labeled samples, caLFDA even managed to outperform LFDA that uses the real and costly ground truth. Consequently, we notice that considering the class as a whole or altering it by decomposing it into

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smaller subclasses does not affect much the efficacy of LFDA. This urged us to review the relevance of the class labels to LFDA and, eventually, to caLFDA.

In fact, the main reason that the LFDA requires a set of class labels is to measure the proximity between the samples, which permits to define LFDA's between-class and within-class scatter matrices, the bedrock to compute LFDA's transformation matrix. Hence, we can consider the set of labeled samples as an adviser that guides LFDA through the process of finding the most discriminating embedding space by telling it to look in one direction or the other. Besides, as precised in [27], LFDA overlooks the powerful and supervisory assistance that class labels provide to a classifier and only exploits them to optimize the global and local structure of the embedding space. In other words, given a set of samples of the same class, LFDA is interested in their locality rather than in how they coherently represent the same meaningful class.

To summarize these facts and observations, what LFDA benefits from the set of labeled samples is a hint about the similar samples in order to preserve their localities and separate them, in the embedding space, from other similar samples.

Seeing LFDA's dependence on the set of class labels from this perspective, we wonder how essential and relevant the expert-made ground truth is to the supervised LFDA and whether it is possible to replace the ground truth by another kind of information. For instance, in the case of an HSI for which we do not dispose of the costly and hard to acquire ground truth, could not we replace this latter by a region map that groups the similar pixels in "classes"? Is the use of the supervised LFDA (and caLFDA) to reduce an HSI only limited to the cases where we know the real land cover classes of the samples?

In order to answer these questions and study the influence of the labeled information on LFDA's performance, in Section III, we exploit the rich spectral and spatial information offered by HSIs to define labeled region maps that we later feed to LFDA instead of the ground truth.

III. Investigation: Influence of the Set of Labeled Samples on LFDA's Performance

In our quest to study the impact of the set of labeled samples on LFDA's efficacy, we propose to replace the costly ground truth by different labeled region maps that do not require an expert nor a prior knowledge of the HSI.

A. Definition of the Labeled Region Maps

We propose to exploit the spectral and/or spatial information of the HSI in order to identify the similar pixels based on their aspects and behaviors and to group them under the same label. Here are our proposed region maps.

1) Using k-Means Based on the Spectral Information: Due to their hundreds of bands, HSIs offer very rich spectral information that we propose to exploit in order to generate a region map where pixels are grouped based on their similar spectral signatures. To do so, we resort to k-means [28] as one of the most famous and commonly used clustering techniques.

Given an HSI, k-means would partition the image's n pixels into k clusters based on the distance between their spectral signatures and on the centroids of the defined clusters. Hence, in the generated clustered image, every cluster would be made of spectrally homogeneous pixels.

Since, in traditional expert-made ground truth, every class would be made of spectrally similar and homogeneous pixels, we consider the labeled samples of the clustered image as our first region map to which we, hereinafter, refer by "clustered."

2) Using the Gridding Based on the Spatial Information: HSIs offer undeniably fruitful spatial information since neighboring pixels share the same spatial context. Hence, we, hereby, propose to generate a region map by simply gridding the image into cells/windows. By doing so, we can choose the cell sizes in advance and ensure not to define larger classes that require a lot of memory to be processed, which is very advantageous.

Since, in traditional expert-made ground truth, every class would be made of samples that share the same spatial context, we consider the labeled samples of the gridded image as our second region map to which we, hereinafter, refer by "gridded."

3) Using the Shearlet Transform Based on the Spatiospectral Information: Since, in the literature, the combination of the HSI's spectral and spatial information proved richer and more informative, we propose to exploit both information to define our third region map that is inspired by the Shearlet-based map of edge-surrounded partitions defined in [29].

To obtain this map, the HSI is first reduced using PCA to keep 98% of its total variance. Then, the reduced image is decomposed using the Shearlet transform. Seeking the edges that do not show much homogeneity with regards to their surroundings, only the finest scale of the obtained decomposition is selected. Next, the coefficients of the kept scale are clustered using k-means into either "edge" or "not-edge." Afterward, the inverse Shearlet transform is applied, and the reconstructed edge bands are fused into one image. This latter is manually thresholded and further enhanced by a morphological cleaning, an erosion, and then a dilation to eliminate at most pointless tiny regions and single pixels. Finally, the image is labeled based on the four-connexity in order to get well-delimited homogeneous regions.

In order to automatically obtain a Shearlet-based region map, we, hereby, propose to replace the manual thresholding step by Otsu's unsupervised thresholding technique [30]. Then, we consider the obtained labeled samples as our third region map to which we, hereinafter, refer by "transformed."

Given that our clustered and gridded region maps, respectively, require the parameters' size of clusters (k) and size of cells, our transformed region map is the only one that is generated in a fully automatic way.

Besides, the clustering and gridding techniques label each and every pixel of the given HSI, which works in their favor as we know the more labeled samples that we feed LFDA, the better it performs. In contrast, due to the automatic thresholding, the transformed region map has the fewest labeled samples. This would probably mislead the reduction process, which, consequently, would yield bad classification results.

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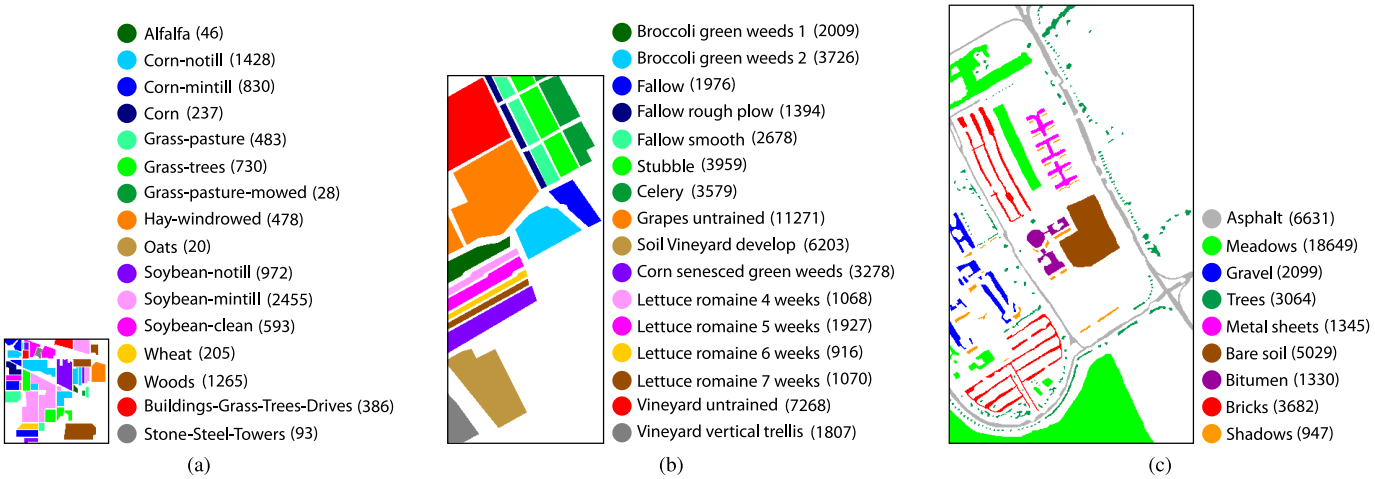


Fig. 2. Ground truth and class labels of the used HSIs, on a scale that shows the difference between their dimensions, with the number of labeled samples per class in brackets. (a) AVIRIS Indian Pines. (b) AVIRIS Salinas. (c) ROSIS Pavia University.

In order to assess the efficiency of these region maps, we propose to use the following experimental protocol.

B. Experimental Protocol

Aiming at studying the influence of the labeled information on LFDA, and at assessing the efficiency of our proposed labeled region maps, we, hereby, compare the results of classifying HSIs reduced by LFDA using the ground truth to those using our region maps. For our experiments, we used the three HSIs[2] illustrated in Fig. 2.

Indian Pines, as illustrated in Fig. 2(a), was captured over the Indian Pines region, IN, USA, and is composed of 145×145 pixels. Salinas, as shown in Fig. 2(b), was acquired over Salinas Valley, CA, USA, and is composed of 512×217 pixels. Both HSIs are acquired by the 224-band Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), represent 16 classes of soil covers, and are reduced to 200 and 204 bands, respectively, to avoid noise and water absorption phenomena. The third HSI, Pavia University, as shown in Fig. 2(c), is captured by the Reflective Optics System Imaging Spectrometer (ROSIS) over Pavia, Italy, is composed of 610×340 pixels and 103 bands, and represents nine classes of land covers.

In our experiments, we reduced these HSIs using LFDA guided by the ground truth and each of our clustered, gridded, and transformed region maps. Then, we used the basic k-nearest neighbors (kNNs) to classify the reduced images. Finally, we assessed the efficiency of our sets of labeled samples based on the classification results measured in terms of average accuracy (AA), overall accuracy (OA), and average reliability (AR). Note that through this article, in each experiment, we used one or multiple of these accuracies in order to have a broader vision of the obtained results and to showcase how the different metrics are coherent.

C. Experimental Results

To give the reader an idea of the nature of our proposed region maps, let us consider Fig. 3 that illustrates the very different-looking labeled region maps of the Indian Pines.

[2]The used hyperspectral data sets are available online at www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes.



Fig. 3. Region maps of Indian Pines. (a) Clustered with $k = 32$. (b) Gridded with 29×29 cells. (e) Transformed.

In fact, based only on the spectral information, the clustered map shows "classes" that are scattered all over the image and that cannot be spatially differentiated. Regarding the gridded map, it blindly divides the image into cells. As for the transformed map, it shows very small and almost meaningless "classes," which is due to Otsu's severe automatic thresholding.

Considering the number of labeled samples in each map, we *a priori* assume that the clustered and gridded region maps would provide LFDA with richer information than the ground truth or the transformed region map, which we believe would give limited information due to their fewer labeled samples.

To verify our thoughts, let us compare the classification results shown in Table I according to different numbers of reduced dimensions. Note that for the configurable clustered and gridded region map, we set the number of clusters to $k = 32$ for Indian Pines and Salinas and to $k = 18$ for Pavia University, and we chose to grid the different images to 10×10 sized cells. Besides, to ensure the fairest comparisons, we randomly selected 10%, 2%, and 2% of the total samples of every class of the ground truth of Indian Pines, Salinas, and Pavia University, respectively, and we used the obtained sets as training samples.

From Table I, we, overall, notice that the obtained accuracies are relatively good with a small number of reduced dimensions and that they get higher when increasing the number of dimensions. Besides, in the case of Indian Pines, we conclude that reducing this HSI by LFDA using the costly ground truth is as efficient as reducing it using one of our region maps. What is more interesting is that the gridded and

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TABLE I

ACCURACIES (AA) OF CLASSIFYING OUR HSIS REDUCED BY LFDA USING THE GROUND TRUTH (GT) AND EACH OF OUR REGION MAPS, ACCORDING TO SEVERAL NUMBERS OF REDUCED DIMENSIONS (RD)

HSI	RD	GT	Clustered	Gridded	Transformed
Indian Pines	1	**34.72**	34.59	34.01	29.32
	4	61.20	57.47	**73.39**	67.89
	7	67.26	62.73	**79.61**	73.41
	10	71.16	65.72	**81.12**	75.29
	15	73.50	67.15	**82.43**	76.98
	20	74.25	68.25	**83.30**	78.08
	25	74.85	70.16	**83.61**	77.54
Salinas	1	53.77	**55.78**	48.93	15.05
	4	88.67	88.82	**89.61**	64.56
	7	91.36	90.79	**93.40**	75.61
	10	91.87	91.29	**94.15**	87.77
	15	92.03	91.63	**94.77**	92.12
	20	92.11	91.72	**95.20**	92.99
	25	92.20	91.80	**95.37**	93.30
Pavia Univ.	1	44	**48**	43	18
	4	**76**	75	74	26
	7	**77**	76	75	42
	10	**77**	**77**	76	47
	15	**78**	77	77	60
	20	78	77	**80**	66
	25	78	77	**81**	68

the transformed region maps manage to guide LFDA better than the ground truth, which explains the higher accuracies they achieve. Furthermore, in the case of Salinas, we got very close and high accuracies that stabilized starting from a very small number of reduced dimensions. In fact, using the gridded map, we, overall, achieved the highest accuracies that are better than the nearly identical accuracies that we got using the clustered region map and the ground truth. However, the transformed map gave the lowest—yet still high—accuracies and was the last to blend with the other maps. Finally, regarding the Pavia University, our clustered and gridded maps proved as efficient in guiding LFDA as the ground truth, while, once again, the transformed region map yielded the worst results.

At this stage, we can confirm that it is possible to use LFDA to reduce an HSI without ground truth and that, for instance, by resorting to one of our proposed automatic and unsupervised labeled region maps. Nevertheless, in order to further test this concept and our proposed labeling methods, we compared the accuracies (in terms of AA) of classifying our HSIs reduced to seven dimensions while varying the number and percentage of randomly selected training samples per class, as reported in Table II. Note that in the case of a small class that counts fewer samples than the chosen number of training samples, the whole class is considered for training.

From Table II, we can once again confirm that reducing an HSI with no ground truth by the supervised LFDA is possible using one of our proposed labeled region maps. However, the latter are not equally efficient. In fact, the clustered map seems to guide LFDA very similar to the ground truth, which we can see in the very close and nearly identical obtained results. Considering the transformed map, it only manages to yield to high accuracies in the case of Indian Pines. However, in the case of the other two HSIs, this region map is the worst to guide LFDA, which explains the worst results that

TABLE II

ACCURACIES (AA) OF CLASSIFYING OUR HSIS REDUCED TO SEVEN DIMENSIONS BY LFDA USING THE GROUND TRUTH (GT) AND EACH OF OUR REGION MAPS, ACCORDING TO SEVERAL NUMBERS AND PERCENTAGES OF TRAINING SAMPLES PER CLASS (TS/C)

HSI	Ts/c	GT	Clustered	Gridded	Transformed
Indian Pines	5	58.22	55.70	**74.16**	67.98
	50	74.73	72.36	**84.93**	81.14
	100	80.29	77.67	**88.61**	84.99
	150	82.34	80.99	**89.90**	86.70
	200	85.39	83.56	**92.19**	88.54
	5%	58.22	55.70	**74.16**	67.98
	10%	67.16	63.04	**82.83**	73.97
	15%	68.46	65.58	**82.39**	78.01
	20%	72.73	69.78	**84.85**	78.44
Salinas	5	80.32	81.12	**89.08**	70.23
	50	91	91.01	**93.46**	77.63
	100	92.66	92.42	**94.18**	80.46
	150	93.17	92.79	**94.40**	80.55
	200	93.80	93.42	**94.94**	81.54
	0.5%	86.89	86.70	**90.72**	69.12
	1%	88.59	89.17	**92.40**	73.98
	1.5%	90.52	90.36	**92.83**	75.80
	2%	91.29	91.11	**93.48**	76.65
	2.5%	90.17	90.31	**93.19**	76.81
	3%	92.07	91.33	**93.95**	77.99
Pavia Univ.	5	65	64	**66**	38
	50	**77**	76	76	48
	100	**81**	80	80	50
	150	**80**	79	79	51
	200	**82**	81	80	53
	0.5%	71	70	**72**	41
	1%	**75**	74	73	42
	1.5%	**74**	**74**	72	42
	2%	**76**	75	74	42
	2.5%	**77**	76	74	43
	3%	**78**	77	75	43

we obtained. Finally, based on the accuracies we got, we can deduce that the gridded region map is the one to guide LFDA in the best way or at least as good as the ground truth.

D. Discussion

To wrap this article, we can confirm that we are able to use the supervised LFDA to reduce an HSI in the absence of its ground truth. To do so, we can replace the costly, rare, and hard to acquire ground truth by a labeled region map that is generated in an automatic and unsupervised way without requiring expertise nor prior knowledge of the classes in the image. For instance, we can generate this map using one of our proposed labeling methods.

Through the experiments we ran on three different HSIs, all our proposed labeled region maps succeeded in guiding LFDA to discriminate the data's classes, and this is reflected in the quite high accuracies we got. What is more interesting is that, in every case we had, at least one of our proposed labeled images performed as good as the ground truth, or even better. This can be explained by the fact that, unlike the ground truth that only labels parts of the image, some of our proposed region maps affect labels to all the data's samples. This way, we are considering the samples that lay between two adjacent classes and which experts tend to discard in the ground truth in order to have the purest and most homogeneous classes possible. However, these samples might contain fruitful information and help discriminate the classes.

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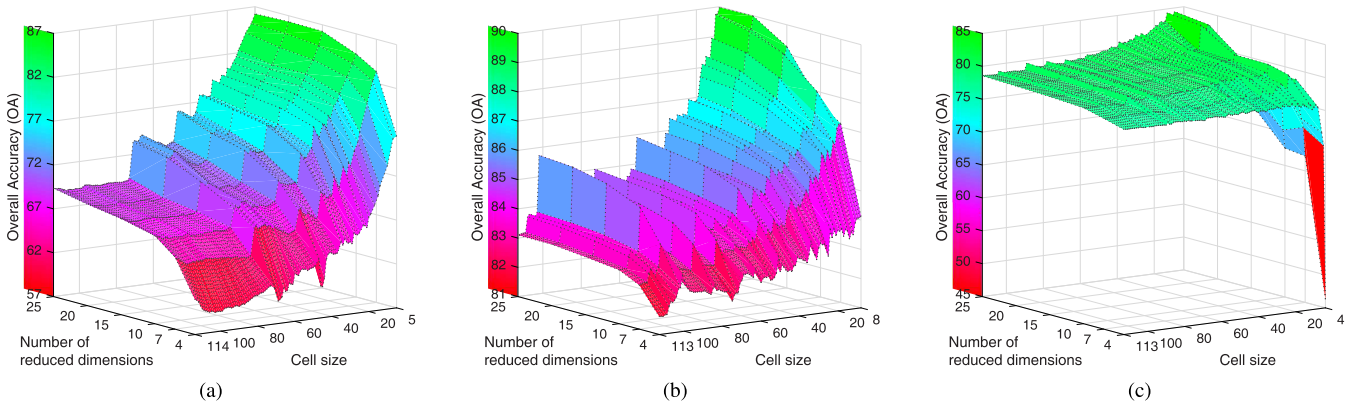


Fig. 4. Accuracies (OA) of classifying (a) Indian Pines, (b) Salinas, and (c) Pavia University reduced by uiaLFDA based on various cell sizes.

Consequently, with our proposed region maps, LFDA avails itself of richer and more global information than the ground truth's narrowed and too specific information.

Aside from the fact that they enable a supervised technique to relatively efficiently reduce an HSI without ground truth, our region maps have a major advantage, which is the generation time. In fact, it takes 10.35, 0.04, and 283.02 s to, respectively, generate a clustered, a gridded, and a transformed region map that is way shorter than the longer time needed to create a ground truth. Hence, we conclude that our proposed region maps are a very good alternative to the ground truth, mainly our gridded map that is the simplest and fastest to be generated and most efficient based on our obtained results.

Note that it would be unfair to compare the execution time of LFDA guided by each of our proposed region maps since this time depends on the number and size of the classes to be processed. However, our region maps do not define the same number nor sizes of classes, which makes LFDA require more or less time depending on the defined classes.

At this stage, we proved it possible to reduce an HSI without a ground truth using LFDA. Next, we propose an unsupervised version of LFDA.

IV. UIALFDA

Considering our study in the previous section of this article, we believe that we have a proof of concept based on which we can define an unsupervised version of LFDA. This latter, which we are calling the uiaLFDA, hereinafter noted uiaLFDA, is dedicated to reducing the abundantly available highly dimensioned HSIs for which we do not have a ground truth and whose labeling process is long and costly.

Our technique is called unsupervised since it works around the supervised aspect of LFDA and overcomes its dependence on the expert-made class labels of the image to be reduced by using information generated in an automatic, blind, and expert-free way. Besides, uiaLFDA is called image-adapted since it replaces the supervised and costly ground truth by the information it extracts from the image. Particularly, we propose to use the gridding technique that labels each and every pixel in the image by simply gridding this latter based on the spatial neighborhood. Since we cannot apply this method to any other type of data except for an image, we call the uiaLFDA image-adapted.

To put it in a nutshell, our proposed uiaLFDA defines its own set of labeled samples by gridding the image into cells of a configurable size and then processes its defined "classes" in the same fashion of LFDA by maximizing the between-class variance and minimizing the within-class variance while preserving the local structure of the data.

Since the size of the defined classes might affect uiaLFDA's efficiency, in the following, we focus on studying its variation and influence on the quality of the reduced features.

A. Tuning the Size of the Cells (Classes)

To ensure a RAM-friendly execution regardless of the used machine, the cell size that uiaLFDA adopts to grid the image has to respect the conditions set by caLFDA. In fact, the gridded region map of uiaLFDA should not define classes/cells containing less than eight pixels, nor more than caLFDA's maximum number of samples per class $n_{\max}$.

According to the formula defined in [14], we set the parameter $n_{\max}$ to 13 000 for Indian Pines and to 12 800 for each of Salinas and Pavia University. Therefore, we varied the cell sizes in $[5, 114]\backslash\{6, 8, 9, 11, 12, 13, 16, 18, 24, 36, 48, 72\}$, in $[8, 113]$, and in $[4, 113]\backslash\{7, 21\}$ for the respective cases of Indian Pines, Salinas, and Pavia University. For more details on how to set $n_{\max}$ according to the data and to the available RAM on the used hardware, the reader can refer to [14].

Fig. 4 illustrates the results (in terms of OA) of classifying our HSIs reduced to various dimensions using uiaLFDA based on the aforementioned cell sizes and using training sets made of randomly selected samples representing 10%, 2%, and 2% of the total samples of every class of the ground truth in the respective cases of Indian Pines, Salinas, and Pavia University.

From Fig. 4(a) and (b), we conclude that in the case of the AVIRIS HSIs, the smaller the cell size, the higher the obtained accuracies. This difference is mainly important in the case of Indian Pines where the variation of accuracies can get up to 20%. However, in the case of Salinas, the difference only reaches 7% in the worst scenario.

For Pavia University, as shown in Fig. 4(c), even though the smallest possible cell sizes gave the worst accuracies compared with all the tested cell sizes, mainly when reducing the image

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to a smaller number of reduced dimensions, the highest accuracies are obtained with relatively smaller cell size. Besides, once again, the accuracies tend to lower as we enlarge the cell size. However, starting from the sizes that give the best accuracies, this diminution does not surpass 5%.

With regard to the results of Fig. 4, we notice that in the case of the AVIRIS images, the smallest possible cell sizes yielded the best results, which was not the case for Pavia University. This can be explained by the fact that Pavia's spatial resolution, also known as ground sample distance (GSD), is 1.3 m between the centers of two adjacent pixels, which is very small compared with the 3.7 m of Salinas or to the 20 m of Indian Pines. Therefore, in the case of Pavia University, using a 4×4 cell would not contain much information, which could explain why larger cells yielded better results.

Apart from its impact on the classification results, we observed that the used cell size affected the execution time of uiaLFDA. In fact, when gridding the image to larger (respectively smaller) cells, uiaLFDA would try to discriminate a few larger (respectively lots of smaller) classes that require a lot of (respectively little) time and memory.

Hence, considering uiaLFDA's efficiency, whether measured by the accuracies or by the execution time, we deduce that the smaller the cell sizes, the better the obtained results.

As to how to set this parameter, we propose to choose the smallest cell size possible—with regard to caLFDA's conditions—that covers an area of 250 m^2. For Indian Pines, one pixel covers 20 m $\times$ 20 m = 400 m^2. Therefore, we resort to the first possible cell size, that is, 5×5. For Salinas, the first cell size that covers more than 250 m^2 is 5×5. However, the smallest size that would meet caLFDA's conditions is 8×8, that is why we opt for it. Finally, for Pavia University, we choose a size of 13×13 since it would ensure a coverage of 1.3 m $\times$ 1.3 m $\times$ 13 $\times$ 13 = 285,61 m^2.

In order to verify how our chosen cell sizes behave and affect uiaLFDA's efficiency, let us consider Table III that compares the classification results (in terms of OA) that we get using uiaLFDA and our aforementioned chosen cell sizes—namely, 5×5, 8×8 and 13×13 for the respective cases of Indian Pines, Salinas, and Pavia University—to those we get using LFDA guided by the ground truth, and to the minimum and maximum accuracies that we get using all possible cell sizes.

From Table III, we notice that with regards to the accuracies, we get using all possible cell sizes or using LFDA, by adopting our chosen cell sizes, uiaLFDA, generally ensures the highest accuracies while mostly outperforming LFDA.

Having defined the parameters of uiaLFDA, it is crucial to compare it to the state-of-the-art unsupervised techniques.

B. Experimental Validation

To further assess its efficiency, we compare uiaLFDA to existing unsupervised feature extraction techniques, namely, PCA [20], the independent component analysis (ICA) [31], LPP [25], the neighborhood preserving embedding (NPE) [32], and the superpixelwise PCA (superPCA) [33].

TABLE III

IMPACT OF THE CHOSEN CELL SIZES ON UIALFDA'S PERFORMANCE (IN TERMS OF OA) WITH REGARDS TO THE MINIMUM AND MAXIMUM OA THAT WE GET USING ALL POSSIBLE CELL SIZES AND TO THE OA THAT WE GET USING LFDA, ACCORDING TO SEVERAL REDUCED DIMENSIONS (RD)

HSI	RD	uiaLFDA	Min	Max	LFDA
Indian Pines	1	41.28	39.25	**41.88**	40.92
	4	**76.59**	59.18	**76.59**	63.83
	7	**83.33**	64.01	**83.33**	68.52
	10	**84.71**	66.44	**84.71**	72.84
	15	**86.25**	67.30	**86.25**	75.54
	20	**86.32**	67.93	**86.32**	76.38
	25	**86.18**	68.86	**86.18**	77.37
Salinas	1	51.31	50.16	**60.33**	58.73
	4	**84.13**	81.58	**84.13**	81.76
	7	**87.12**	82.48	**87.12**	83.84
	10	87.60	82.83	**88.05**	84.84
	15	**89.26**	82.98	**89.26**	85.15
	20	**89.99**	83.08	**89.99**	85.29
	25	**89.98**	83.10	**89.98**	85.36
Pavia Univ.	1	49.78	44.72	52.08	**55.44**
	4	76.41	46.32	76.60	**77.30**
	7	**79.89**	67.25	**79.89**	79.46
	10	**80.70**	67.62	**80.70**	80.36
	15	80.77	73.04	80.77	**81.06**
	20	**83.87**	73.78	**83.87**	80.91
	25	84.41	78.23	**84.44**	80.60

TABLE IV

ACCURACIES (AA) OF CLASSIFYING OUR HSIS REDUCED BY UIALFDA AND EXISTENT TECHNIQUES TO SEVERAL REDUCED DIMENSIONS (RD)

Data sets	RD	PCA	ICA	LPP	NPE	superPCA	uiaLFDA
Indian Pines	1	65.13	16.26	62.09	20.24	**97.50**	85.95
	4	64.41	15.95	54.53	59.49	**97**	77.23
	7	67.29	16.95	59.16	61.77	**97.38**	84.60
	10	65.13	18.63	62.09	61.54	**97.50**	85.95
	15	66.85	20.53	61.47	61.02	**97.50**	87.28
	20	67.04	20.57	61.81	61.18	**97.57**	87.40
	25	66.27	23.14	62.18	58.96	**97.57**	87.16
Salinas	1	**54.74**	9.290	40.53	9.08	42.54	48.23
	4	91.14	10.93	90.29	9.42	**97.86**	91.91
	7	92.21	15.33	93.57	9.50	**98.16**	94.19
	10	92.29	22.09	93.82	9.78	**98.23**	94.84
	15	92.19	34.72	94.72	11.63	**98.25**	95.26
	20	92.20	43.56	94.39	12.17	**98.26**	95.35
	25	92.27	54.65	93.11	12.79	**98.26**	95.36
Pavia Univ.	1	44	14	38	38	32	**46**
	4	76	14	77	73	**85**	**85**
	7	77	14	77	75	**88**	**88**
	10	78	14	74	72	**89**	88
	15	79	15	70	69	**89**	88
	20	79	15	63	63	**89**	88
	25	79	16	57	58	**89**	88

Table IV reports the results (in terms of AA) of classifying our HSIs reduced to several dimensions by the aforementioned techniques, using the same sets of training samples made of 10%, 2%, and 2% of every class in the respective ground truth of Indian Pines, Salinas, and Pavia University.

As reported in Table IV, our proposed uiaLFDA always outperforms PCA (except for one case), LPP, ICA, and NPE. Besides, the latter two give the worst accuracies, while the best accuracies are mostly achieved by superPCA that outperforms uiaLFDA by an average of 12.4% and 3.7% in the respective cases of Indian Pines and Salinas. However, in the case of Pavia University, uiaLFDA outperforms superPCA when reducing the HSIs to very small numbers of reduced dimensions. Otherwise, they both yield very close accuracies.

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TABLE V

ACCURACIES (AR) OF CLASSIFYING OUR HSIS REDUCED BY UIALFDA, PCA, LPP, AND SUPERPCA TO SEVEN DIMENSIONS, ACCORDING TO DIFFERENT NUMBERS AND PERCENTAGES OF TRAINING SAMPLES PER CLASS (TS/C)

HSI	Ts/c	PCA	LPP	superPCA	uiaLFDA
Indian Pines	5	44.57	36.90	26.22	**56.81**
	50	61.71	55.25	69.48	**78.42**
	100	66.01	60.26	**87.82**	83.51
	150	73.03	66.80	**93.77**	85.66
	200	77.21	71.15	**96.54**	87.94
	5%	61.62	54	78.89	**83.83**
	10%	69.17	61.51	**90.57**	85.24
	15%	72.74	64.20	**97.37**	87.51
	20%	73.41	67.21	**97.39**	88.88
Salinas	5	82.83	88.25	36.44	**89.69**
	50	89.68	91.32	69.85	**92.60**
	100	90.34	92.63	83.08	**93.27**
	150	91.72	92.95	87.33	**94.19**
	200	92.24	93.69	89.63	**94.44**
	0.5	87.85	**91.54**	57.19	90.95
	1%	90.76	92.74	68.04	**93.42**
	1.5%	91.54	92.54	77.78	**93.55**
	2%	91.56	93.47	81.64	**93.96**
	2.5%	91.53	93.56	87.31	**94.10**
	3%	92.11	93.45	89.25	**94.21**
Pavia Univ.	5	64	61.75	41.24	**69.87**
	50	71.43	70.35	61.51	**81.23**
	100	74.16	71.55	69.39	**82.95**
	150	76.79	73.77	74.67	**84.87**
	200	78.02	75.01	76.43	**86.90**
	0.5	74.58	72.58	62.17	**86.14**
	1%	75.99	75.52	70.06	**86.83**
	1.5%	78.16	76.19	75.66	**87.02**
	2%	78.33	77.26	79.20	**87.81**
	2.5%	79.08	77.56	81.65	**88.55**
	3%	81.09	79.68	84.90	**89.65**

The better performance of superPCA might be explained by the fact that unlike PCA that processes a given HSI as a whole, this technique exploits the spatial information of the HSI and processes it based on its superpixels. To do so, it 1) performs a superpixel segmentation on the first principal component of the HSI in order to obtain homogeneous regions, applies PCA to each of the obtained regions, and rearranges and combines the reduced superpixels to form the reduced image. It is worth pointing out that superPCA reduces a given image by analyzing its reduced version.

Let us call the result of superPCA segmenting the first principal component of the HSI a region map. The number of superpixels in this map is configurable, and its optimal value, which we used by default in Table IV, is 100.

If we are to consider superPCA and uiaLFDA as techniques that analyze a region map, it is fairer to compare them using maps made of the same number of regions. Therefore, to meet the number of regions in the gridded region maps of uiaLFDA, in the rest of this article, we set the number of superpixels of superPCA to 841, 1792, and 1269 in the respective cases of Indian Pines, Salinas, and Pavia University.

Table V compares the results (in terms of AR) of classifying our HSIs reduced to seven dimensions by uiaLFDA and by the state-of-the-art techniques that compete with it, namely, PCA, LPP, and superPCA, using the aforementioned numbers of superpixels per image, while varying the number and percentage of randomly selected training samples per class.

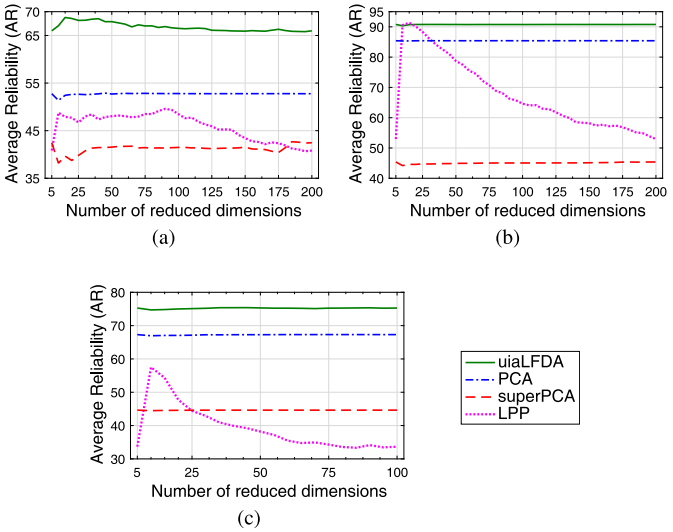


Fig. 5. Accuracies (AR) of classifying (a) Indian Pines, (b) Salinas, and (c) Pavia University when increasing the number of their reduced dimensions and only using ten training samples per class for classification.

As stated in Table V, we notice that by modifying the number of superpixels of superPCA, this technique seems less efficient, mainly when reducing the HSIs to a smaller number of dimensions. However, superPCA still manages to be the most reliable in the case of Indian Pines and to outperform uiaLFDA with an average of 7.45%. In contrast, in the cases of Salinas and Pavia University, uiaLFDA outperforms superPCA with respective averages of 17.89% and 14.09% while nearly always being the most reliable. In fact, PCA and LPP outperform superPCA in the case of Salinas, while all three give very close results in the case of Pavia University.

The curse of dimensionality, also known as Hughes phenomenon [34], results in decreasing classification results when increasing the number of features in the presence of very few labeled samples. To evaluate how robust are uiaLFDA's extracted features to this curse, we used the same ten training samples per class to classify our HSIs while increasing the number of their dimensions, as illustrated (in terms of AR) in Fig. 5.

According to Fig. 5, PCA, superPCA, and uiaLFDA proved very stable and did not confuse the classifier as we increased the number of extracted features, unlike LPP that caused the accuracies to considerably decrease. Besides, once again, uiaLFDA proved to always be the most reliable and to yield the best results even in the presence of very few training samples.

Table VI reports the accuracies per class of classifying our HSIs reduced to seven dimensions, using 50 training samples per class. We conclude that, overall speaking, uiaLFDA outperforms the state-of-the-art methods. In fact, for Indian Pines, uiaLFDA gives the best accuracies for 11 classes and is second to superPCA for the other five classes. For Salinas, it also gives the best results for 11 classes while being outperformed in the left five classes by either PCA or LPP with a gap of 6% in the worst case. As for Pavia University, uiaLFDA gives the best accuracies for seven classes and is outperformed by less than 2% in the left two classes by either PCA or superPCA. Finally, uiaLFDA gets the best AA, OA, and AR except for the OA of

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TABLE VI

ACCURACIES OF CLASSIFYING OUR HSIS, REDUCED EACH TO SEVEN DIMENSIONS, USING TRAINING SETS MADE OF 50 SAMPLES PER CLASS

AVIRIS Indian Pines					AVIRIS Salinas					ROSIS Pavia University				
Class	PCA	LPP	superPCA	uiaLFDA	Class	PCA	LPP	superPCA	uiaLFDA	Class	PCA	LPP	superPCA	uiaLFDA
1	**100**	**100**	**100**	**100**	1	98.31	99.45	89.70	**99.50**	1	68.42	67.11	67.05	**76.69**
2	44.26	42.44	60.99	**67.65**	2	**99.33**	98.87	84.17	98.77	2	66.26	73.08	50.53	**85.18**
3	56.75	41.20	**79.76**	72.41	3	98.48	91.70	90.69	**99.19**	3	66.27	68.65	75.80	**87.28**
4	67.93	62.03	**96.62**	84.81	4	99.57	**99.71**	99.21	**99.71**	4	92.17	92.04	54.63	**93.64**
5	84.47	79.71	86.54	**89.65**	5	96.08	92.91	87.15	**97.24**	5	98.96	**99.63**	84.46	**99.63**
6	87.12	74.11	81.92	**98.08**	6	99.01	**99.62**	86.11	99.42	6	62.06	66.61	66.36	**85.01**
7	**100**	**100**	**100**	**100**	7	99.05	99.30	77.06	**99.36**	7	81.58	75.26	**98.12**	94.29
8	89.54	83.47	**99.58**	94.35	8	67.16	**73.17**	46.44	67.11	8	71.51	61.19	71.40	**71.56**
9	**100**	**100**	**100**	**100**	9	95.12	95.58	64.55	**97.19**	9	**99.89**	99.79	99.68	99.68
10	67.80	58.13	73.56	**78.91**	10	85.69	89.35	78.43	**90.70**					
11	45.91	40.33	54.18	**57.88**	11	93.54	95.04	90.54	**96.35**					
12	54.47	47.72	83.98	**90.56**	12	99.90	99.38	86.51	**99.95**					
13	91.22	91.22	**99.51**	**99.51**	13	98.25	**99.24**	98.36	98.80					
14	73.75	71.62	71.94	**82.53**	14	96.26	**99.16**	90	96.92					
15	41.45	33.94	**97.41**	72.28	15	55.31	63.86	50.69	**66.94**					
16	97.85	95.70	**100**	97.85	16	91.53	96.46	73.82	**98.51**					
AA	75.16	70.10	86.62	**86.65**	AA	92.04	93.30	80.84	**94.10**	AA	79	78	74	**88**
OA	61.57	55.27	72.92	**76.15**	OA	84.71	**87.31**	70.09	87.26	OA	70.66	73.01	61.92	**84.44**
AR	61.71	55.25	69.48	**78.42**	AR	89.68	91.32	69.85	**92.60**	AR	71.43	70.35	61.51	**81.23**

TABLE VII

EXECUTION TIMES IN SECOND OF PCA, LPP, SUPERPCA, AND UIALFDA WHEN REDUCING OUR HSIS TO SEVEN DIMENSIONS

Data sets	PCA	LPP	superPCA	uiaLFDA
Indian Pines	**2.76**	143.83	2.90	3.01
Salinas	14.43	4466.40	**8.51**	16.40
Pavia Univ.	7.49	16611.56	**6.16**	10.69

Salinas where LPP is best since it yields better classification of the larger Grapes Untrained class (number 8).

Table VII compares the execution times of uiaLFDA and the existent methods. Even though it is not the fastest, uiaLFDA does not take as much time as LPP and is slower than PCA and superPCA by only an average of 4 s, which is, in our humble opinion, tolerable given the better accuracies it yields.

V. CONCLUSION

Traditionally, to process an HSI, supervised methods, such as LFDA, require the highly costly, hard to acquire, and rarely available ground truth, which affects class labels to generally very few of the image's samples. Moreover, these techniques necessitate a big number of labeled samples in order to perform well, which is not always affordable with HSIs.

In this article, we questioned the relevance of the ground-truth's labeled samples to LFDA and studied to which extent this latter depends on them by replacing the ground truth with three labeled region maps that are automatically generated based on the HSI's spatial and/or spectral information, without requiring an expert nor prior knowledge of the image. Each of the maps results from one of these procedures: 1) clustering the image by k-means; 2) gridding the image into cells; and 3) transforming the image using Shearlet transform.

Given the accuracies of classifying three known HSIs reduced by LFDA guided by our region maps instead of the ground truth, we proved that this latter is not indispensable to LFDA. We also proved that LFDA does not work only in the presence of expert-made labeled samples, which in the context of HSIs processing are very costly to provide.

Based on these results, we defined the uiaLFDA, which is able to reduce an HSI whose ground truth is unavailable by replacing this latter with an expert-free labeled region map. To do so, given the HSI's spatial resolution, uiaLFDA grids the image into the smallest cells that: 1) contain more than eight samples; 2) cover at least 250 m^2; and 3) contain less than n_{max} samples, with this latter parameter, defined in [14], ensuring that uiaLFDA would not require more memory than the RAM of the used hardware.

Through our experiments ran on three HSIs, we proved that uiaLFDA is an efficient extension of the supervised LFDA to reduce HSIs without ground truth. Actually, in most cases, we proved that uiaLFDA even outperforms the original LFDA that uses the costly ground truth. Besides, compared with the state-of-the-art methods, uiaLFDA almost always outperforms the existent unsupervised PCA, ICA, LPP, and NPE. Finally, compared with the very recent superPCA that applies PCA to the superpixels that it extracts from the first principal component resulting from reducing the HSI by PCA, uiaLFDA not only exploits a very simple and basic gridded region map but also gives very close results that, in most cases, are better than those of superPCA.

As a future research line, we intend to study whether using an automatic and expert-free set of labeled samples could extend efficient supervised dimensionality reduction techniques to reduce HSIs without ground truth.

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